Abhishek Hegade K R

Contact Information

E-mail: ah4278@princeton.edu, Personal website.

Academic Appointments

Associate Research Scholar, Princeton Gravity Initiative 2025–present

Education

PhD in Physics, University of Illinois Urbana-Champaign 2020–2025
Advisor: Nicolás Yunes

BSc (Hons.) in Mathematics and Physics, Chennai Mathematical Institute 2017–2020
Advisors: K. G. Arun and Alok Laddha
Gold medal for graduating at the top of my class.

Research Interests

Relativistic astrophysics, gravitational-wave source modeling, neutron-star tidal dynamics, extreme-mass-ratio inspirals in accretion disks, black-hole accretion, and relativistic plasma physics.

Publications

I have 20 published, accepted, or submitted papers: 14 as first author or equal contribution and 6 as second author. Student-led papers are marked with [s], equal contributions with [=], and Editor's Suggestions with [†]. Four additional manuscripts are in preparation. Google Scholar: 440 citations, h-index 12.

Published or accepted:

1. **Abhishek Hegade K. R.**, James M. Stone, "Kinetic magnetohydrodynamics and Landau fluid closure in relativity," *J. Plasma Phys.*, accepted (2026). [arXiv]
2. [s]Anand Balivada, **Abhishek Hegade K. R.**, Nicolás Yunes, "A New Spin on Dissipative Tides: First-Post-Newtonian Effects in Compact Binary Inspirals," *Phys. Rev. D* **114**, 084028, (2026). [published] [arXiv]
3. **Abhishek Hegade K. R.**, K.J. Kwon, Tejaswi Venumadhav, Hang Yu, Nicolás Yunes, "Relativistic mode sums for neutron-star tidal response," *Phys. Rev. D* **114**, 064048 (2026). [published] [arXiv]
4. **Abhishek Hegade K. R.**, Yumu Yang, Mauricio Hippert, Jacquelyn Noronha-Hostler, Jorge Noronha, Nicolás Yunes, "Dynamical tidal response of neutron stars as a probe of dense-matter properties," *Phys. Rev. D* **114**, 044014 (2026). [published] [arXiv]
5. [s]Jayana A. Saes, **Abhishek Hegade K. R.**, Nicolás Yunes, "Universal relations with dynamical tides," *Phys. Rev. D* **113**, 104045 (2026). [published] [arXiv]
6. **Abhishek Hegade K. R.**, Charles F. Gammie, Nicolás Yunes, "Relativistic treatment of accretion disk torques on extreme mass-ratio inspirals around spinning black holes," *Phys. Rev. D* **112**, 124068 (2025). [published] [arXiv]
7. **Abhishek Hegade K. R.**, Charles F. Gammie, Nicolás Yunes, "Relativistic treatment of accretion disk torques on extreme mass-ratio inspirals around nonspinning black holes," *Phys. Rev. D* **112**, 124012 (2025). [published] [arXiv]
8. [†]**Abhishek Hegade K. R.**, K.J. Kwon, Tejaswi Venumadhav, Hang Yu, Nicolás Yunes, "Relativistic and Dynamical Love Numbers," *Phys. Rev. Lett.* **136**, 071401 (2026). [published] [arXiv]
9. **Abhishek Hegade K. R.**, Justin L. Ripley, Nicolás Yunes, "Dissipative tidal effects to next-to-leading order and constraints on the dissipative tidal deformability using gravitational wave data," *Phys. Rev. D* **110**, 044041 (2024). [published] [arXiv]

10. Justin L. Ripley, **Abhishek Hegade K. R.**, Rohit S. Chandramouli, Nicolás Yunes, "A constraint on the dissipative tidal deformability of neutron stars," *Nat. Astron.* **8**, 1277–1283 (2024). [published] [arXiv]
11. **Abhishek Hegade K. R.**, Justin L. Ripley, Nicolás Yunes, "Dynamical tidal response of non-rotating relativistic stars," *Phys. Rev. D* **109**, 104064 (2024). [published] [arXiv]
12. Justin L. Ripley, **Abhishek Hegade K. R.**, Nicolás Yunes, "Probing internal dissipative processes of neutron stars with gravitational waves during the inspiral of neutron star binaries," *Phys. Rev. D* **108**, 103037 (2023). [published] [arXiv]
13. **Abhishek Hegade K. R.**, Justin L. Ripley, Nicolás Yunes, "The non-relativistic limit of first-order relativistic viscous fluids," *Phys. Rev. D* **107**, 124029 (2023). [published] [arXiv]
14. **Abhishek Hegade K. R.**, Elias R. Most, Jorge Noronha, Helvi Witek, Nicolás Yunes, "How Do Axisymmetric Black Holes Grow Monopole and Dipole Hair?" *Phys. Rev. D* **107**, 104047 (2023). [published] [arXiv]
15. **Abhishek Hegade K. R.**, Justin L. Ripley, Nicolás Yunes, "Where and why does Einstein-scalar-Gauss-Bonnet theory break down?" *Phys. Rev. D* **107**, 044044 (2023). [published] [arXiv]
16. **Abhishek Hegade K. R.**, Elias R. Most, Jorge Noronha, Helvi Witek, Nicolás Yunes, "How do spherical black holes grow monopole hair?" *Phys. Rev. D* **105**, 064041 (2022). [published] [arXiv]
17. †Deep Chatterjee, **Abhishek Hegade K. R.**, Gilbert Holder, Daniel E. Holz, Scott Perkins, Kent Yagi, Nicolás Yunes, "Cosmology with Love: Measuring the Hubble constant using neutron star universal relations," *Phys. Rev. D* **104**, 083528 (2021). [published] [arXiv].

In preparation:

1. **Abhishek Hegade K. R.**, James M. Stone, "Kinetic Physics of Black Hole Accretion with Landau-Fluid Closures".
2. =**Abhishek Hegade K. R.**, Alexander J. Dittmann, Callum W. Fairbairn, "Eccentric Extreme-Mass-Ratio Inspirals in Gaseous Disks".
3. sFangyuan Yu, **Abhishek Hegade K. R.**, Sylvia Biscoveanu, "Join inference of the equation of state, cosmology and the mass-spectrum of neutron stars".
4. =Jacob Fields, **Abhishek Hegade K. R.**"Nonlinear tidal interactions from numerical relativity".

Under review:

1. =Alexander J. Dittmann, **Abhishek Hegade K. R.**, Callum W. Fairbairn, "Extreme-Mass-Ratio Inspirals in Gaseous Disks", submitted to PRD.
2. **Abhishek Hegade K. R.**, Chris Hamilton, "Swing amplification in star-gas disks", submitted to ApJ.
3. K.J. Kwon, **Abhishek Hegade K. R.**, Tejaswi Venumadhav, Hang Yu, Nicolás Yunes, "Nonlinear Perturbations of Relativistic Stars", submitted to PRD.

Talks and Workshops

Invited

1.	Theory of Tidal Deformation: From Nuclear Theory to Gravitational Wave Observations, ICTS, Bengaluru, India.	November 2026
2.	PCTS Lunch Talk, Princeton University, Princeton, NJ.	November 2026
3.	Journal of Plasma Physics Colloquium (online).	October 2026
4.	TAPIR Seminar, Caltech, Pasadena, CA.	October 2026
5.	Astro Lunch Talk, UC Santa Barbara, Santa Barbara, CA.	October 2026
6.	Simons Meeting on Black Holes and Strong Gravity, New York, NY.	May 2026
7.	Gravity Seminar, Penn State University, State College, PA.	April 2026
8.	SISSA Gravitational-Wave Group Meeting (online).	December 2025
9.	Princeton Gravity Initiative Seminar, Princeton University, Princeton, NJ.	September 2025
10.	From Colliders to the Cosmos, Institute for Nuclear Theory, Seattle, WA.	September 2025
11.	Gravitational Physics Seminar, IIT Gandhinagar (online).	July 2025
12.	Gravity Seminar, University of Illinois Urbana-Champaign, Urbana, IL.	May 2025

13. GRITTS Seminar, MIT, Cambridge, MA. October 2024
14. Emanuele Berti Group Meeting, Johns Hopkins University, Baltimore, MD. October 2024
15. ICTS Astrophysics Seminar, ICTS, Bengaluru, India. January 2024
16. VandyGRAF Seminar, Vanderbilt University, Nashville, TN. November 2023
17. Advances in Chern-Simons Classical and Quantum Gravity, ICERM, RI. May 2022

Selected Contributed

1. Leinweber Forum for Theoretical Physics, Princeton, NJ. October 2026
2. Elias Most Group Meeting, Caltech, Pasadena, CA. May 2026
3. Black Holes from Theory to Observations, Institute for Advanced Study, NJ. May 2026
4. APS April Meeting, Denver, CO April 2026
5. APS April Meeting, Anaheim, CA. April 2025
6. APS April Meeting, Sacramento, CA. April 2024
7. APS April Meeting, Minneapolis, MN. April 2023
8. APS April Meeting, New York, NY. April 2022
9. Midwest Relativity Meeting, Urbana-Champaign, IL. November 2021

Awards

Illinois Center for Advanced Studies of the Universe - Physics Fellowship Fall 2024 – Spring 2025
Scott Anderson Award, University of Illinois Urbana Champaign Spring 2023
University Fellowship, University of Illinois Urbana Champaign Fall 2022
Medal of Excellence, Chennai Mathematical Institute 2020
Tuition fee waiver, Chennai Mathematical Institute 2017–2020
KVPY Fellow, Department of Science and Technology of the Government of India 2017–2020
Indian Academy of Sciences, Summer Research Fellowship Summer 2019
National Talent Search Examination - State Scholar 2015

Teaching Experience

Classical Mechanics I Fall 2019
Teaching assistant and grader.

PHYS 402: Light Fall 2020, Spring 2021, Spring 2022
Led discussion and lab sections; served as a grader.

PHYS 436: Electrodynamic Fields II Spring 2023
Led discussion sections and served as a grader.

Research Mentoring

Jason Yu (advisor: Sylvia Biscoveanu) 2026–present
Mentor for a graduate research project; Preparing work for publication.

Taeho Kim (advisor: Sylvia Biscoveanu) 2025–2026
Research mentor for a Princeton undergraduate senior thesis.

Jayana A. Saes (advisor: Nicolás Yunes) 2024–present
Graduate research mentor. One publication and currently mentoring ongoing research.

Anand Balivada (advisor: Nicolás Yunes) 2024–present
Graduate research mentor. One publication and currently mentoring ongoing research.

Leadership and Service

Co-organizer: Princeton Gravity Initiative Lunch Seminar 2025–present
Co-organizer: Princeton University–IAS Gravitational-Waves Meeting 2025–present
Co-organizer: LFTP Workshop, "Binary Black Holes in Their Natural Habitats" 2026
Core Member: LISA Collaboration Astrophysics Working Group 2026–present

Referee: ApJ, PRD, PRL, JCAP